GEOG 654: Final Project Proposal
Daylan Salmeron

Pantanal on Fire: Predicting Wildfires in the Brazilian Pantanal

Research Problem

The Brazilian Pantanal is the largest tropical wetland in the world. It is home to a vast variety of species such as Jaguars, macaws, giant river otters, and many other endangered species. In 2020 about 29% of it was burned due to record-breaking wildfires (Kumar et al., 2022), and surveys have estimated that roughly 17 million vertebrates were killed directly as a result (Tomas et al., 2021).

Given its large extent, fire does not fall evenly across the biome. Thus, knowing which places are most likely to burn in a month based on previous geospatial patterns, could help identify which areas need the most attention. This project aims to build and evaluate such a model.

This study also serves a wider purpose. It provides an opportunity to explore how satellite fire records, gridded weather, flooding and land cover interact across different resolutions and time steps. This opens the door to later ask how such events affect the people living in nearby areas.

Background

The Pantanal is a well-studied area in Brazil, and the 2020 wildfires drew considerable attention from the academic community. Kumar et al. (2022) used a Random Forest to assess what drove the 2020 fires and found that drought alone could not explain where the region burned. Land use, however, played an important role, as the study found that 52% of protected areas burned, against 0.5% of areas with high cattle density. While drought set the conditions, land use appeared to influence fire spread. That said, this study focused on explaining an event that had already occurred rather than predicting one in advance.

Predictive work on the Pantanal does exist. Viganó et al. (2018) ran a model on monthly fire counts in Corumbá, a city locally known as the capital of the Pantanal, using regression and ARIMA[1] on weather variables. Similarly, Narciso et al. (2025) developed Saripan, a web-based system that forecasts fire risk for each Pantanal municipality up to three days in advance. Both studies predict how much fire to expect in a large administrative unit over a short horizon, rather than mapping which locations within it are most likely to burn. Spatial susceptibility mapping has been applied to areas nearby. Bustillo Sanchez et al. (2021) used a Random Forest in Santa Cruz, Bolivia. They validated their model with a spatial block holdout and a temporal test on later years, though only at annual resolution. This project sits between these two approaches, using spatial prediction at 5 km, at monthly steps, with only conditions known before the next month begins.

Objectives

- Build a dataset of fire occurrence in the Pantanal at 5 km, by month from 2019 to 2025.
- Predict whether a cell will burn in a given month using only conditions known before that month begins.
- Test the model on 2025, a year held out from fitting, tuning and calibration.
- Compare it against simple alternatives, such as where fire has usually occurred before.
- Test whether the conclusions hold under different data preparation choices.

[1] ARIMA: Autoregressive Integrated Moving Average.

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Study Area

The study area for this project is the Brazilian Pantanal, as mapped by the IBGE[2], covering 150,880 km^2 across the states of Mato Grosso and Mato Grosso do Sul. It was divided into 6,584 cells of 5 km on a side, cut to the boundary, a size that matches the resolution of the weather data. The Pantanal suits this study because fire activity is strongly concentrated between August and October.

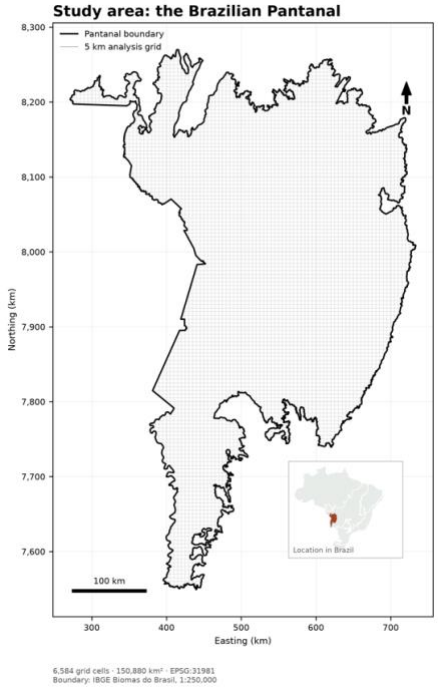


Figure 1. The Brazilian Pantanal and the 5 km analysis grid.

Data

Dataset	Source	Type and resolution	Processing
Brazilian Biomes	IBGE	Vector, 1:250,000	Filtered to Pantanal, reprojected
VIIRS active fire, NOAA-20	NASA FIRMS	Points, 375 m, 2019-2025	Non-vegetation types dropped, counted per cell and month
CHIRPS rainfall and CHIRTS-ERA5 temperature	Climate Hazards Center	Raster, 0.05°, monthly	Sampled at each cell's representative point
MapBiomas land cover and surface water	MapBiomas	Raster, 30 m, yearly and monthly	Percentage of each cell by class
Roads and waterways	OpenStreetMap	Vector lines, 2020	Distance from each cell to the nearest line

All datasets are publicly available and have been reprojected from their native CRS, SIRGAS 2000 for IBGE and WGS 84 for the rest, to EPSG:31981, so areas and distances are in meters. Fire detections come from the VIIRS 375 m active fire product (Schroeder et al., 2014), rainfall from CHIRPS (Funk et al., 2015), temperature from CHIRTS-ERA5 (Funk et al., 2019), and land cover from MapBiomas Collection 11 (Souza et al., 2020).

[2] IBGE: The Brazilian Institute of Geography and Statistics (Instituto Brasileiro de Geografia e Estatística)

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Proposed Methodology

The modeling table has one row for each grid cell in each target month, August – October, for every year from 2019 to 2025. The outcome is whether a fire has been detected, at least once, in that cell during that month.

Every predictor describes the conditions before the predicted month, so that the model only uses what would be known in advance. Rainfall is summed over the previous month, the previous three months and the previous wet season. The wet season is defined as November to March, based on the monthly rainfall pattern. Maximum temperature is taken as the previous month's value and the average of the previous three. Flooding comes from the previous months, and land cover from the previous year. Roads and waterways do not generally change drastically over the study period, so they are taken from a 2020 OpenStreetMap snapshot.

A Random Forest is then trained on 2019 to 2024 and used to predict 2025.

Model Validation

Entire blocks of neighboring cells are held out at a time[3], so the model cannot score well by memorizing nearby cells. Each training year is also predicted using the others, while 2025 is held out from fitting, tuning, threshold choice and calibration and used only for the final test. Results are then compared against two simple alternatives: how often each cell has burned before, and whether it burned the previous month.

Fire is not a common event, so getting most cells right does not provide much insight. Instead, the model is judged on whether it puts the right places near the top of the list, and how many of 2025's real fires fall inside the top 1%, 5% and 10% of the flagged cells.

What is tested for sensitivity?

Choice	Baseline	Also tested
Smallest cell kept	Half a full cell	All cells, quarter, three quarters, whole cells only
Cell size	5 km	10 km
How far back weather is summed	1 and 3 months	1 month only
Size of held-out blocks	50 km	25 km, 75 km
Fire detections used	All	Only the more certain ones

Cutting the grid to the boundary leaves cells of very different sizes, and a small cell can record fewer fires simply because it covers less ground. Choices like this are known to change results (Openshaw, 1984; Fotheringham and Wong, 1991), so the analysis is repeated several ways rather than defending one.

[3] Cells are grouped into 50 km blocks by integer division of their coordinates, and whole blocks are assigned to folds. A random split would leave a test cell's neighbors in the training set, and since fire is spatially clustered the model could score well by recognizing a neighbor rather than by generalizing to a new location.

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